

Spatially Dynamic Mixture-of-Experts for Adverse-Weather Salient Object Detection

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Abstract— Salient object detection (SOD) degrades under adverse weather. Existing approaches condition on explicit weather labels or apply mixture-of-experts (MoE) to image restoration rather than segmentation. We present SpatialMoE-SOD, routing individual spatial tokens to separate expert networks at three independent pyramid scales without using weather labels during training or inference. A depthwise convolutional router and MLP select top- $K=2$ experts from 8 per scale via true sparse dispatch. Per-token routing entropy is injected as an explicit feature channel into the cross-attention decoder via learnable additive fusion. On WXSOD, the model achieves MAE 0.0192 on 1,500 synthetic test images spanning 9 weather categories and 0.0168 on 554 real-world test images spanning 5 weather categories.

Index Terms— salient object detection, mixture-of-experts, adverse weather, sparse routing, cross-attention decoder

I. INTRODUCTION

SOD identifies visually prominent regions as pixel-wise masks, but performance degrades under adverse weather. Two strategies address this: (1) weather-label-conditioned models—NIFM [1]—require metadata at inference; (2) MoE for weather restoration—WM-MoE [2]—targets pixel regression, not segmentation. To our knowledge, existing MoE-SOD methods primarily explore modality-, scale-, or task-level routing for multi-modal inputs rather than per-token routing for single-RGB SOD. To our knowledge, no existing method combines (a) spatial token-level routing, (b) independent routing at multiple pyramid scales, and (c) an SOD decoder informed by routing confidence, all without weather-specific supervision. Our contributions:

- 1) A token-level spatial router that assigns each token to top- $K=2$ experts from 8 per scale, using depthwise-convolutional context, noisy top- K gating, and true sparse dispatch.
- 2) Three independent routers at scales $1/4, 1/8, 1/16$, each with a separate expert pool.
- 3) Routing entropy, normalized by $\log 2$ for $K=2$, is injected as an explicit decoder feature via learnable additive fusion.

II. RELATED WORK

A Adverse-Weather SOD. NIFM [1] fuses noise indicators from explicit weather-type one-hot vectors. WXSOD [3] provides 14,945 images with per-image weather labels and synthetic/real-world test splits for weather-robust SOD.

B MoE for Weather Restoration. WM-MoE [2] uses weather-aware routing for adverse-weather image restoration, whereas our method performs spatial token-level routing for single-RGB SOD without weather-specific supervision. Existing MoE weather methods target pixel regression, not segmentation.

C MoE for SOD. To our knowledge, existing MoE-SOD methods primarily explore modality-, scale-, or task-level routing for

multi-modal inputs rather than per-token routing for single-RGB SOD. V-MoE [4] established token-level top- K routing for classification. Routing entropy has been explored for uncertainty estimation in MoE classifiers. We additionally inject it as an explicit decoder feature for dense prediction, although its isolated contribution has not been evaluated by an ablation.

III. METHOD

Fig. 1 illustrates the full architecture. A shared PVTv2-B4 backbone extracts multi-scale features; three independent SpatialMoE layers process tokens at each scale; a cross-attention decoder fuses the results and produces dense predictions. We detail each component below.

A Multi-Scale Features. A PVTv2-B4 backbone [5] pre-trained on ImageNet extracts features at three scales via $\text{out_indices} = (0, 1, 2)$. Each scale is projected to $C=256$ via 1×1 convolution:

$$\mathbf{X}_s = \text{Proj}_s(\text{Backbone}(\mathbf{X})_s) \in \mathbb{R}^{C \times H_s \times W_s}, \quad (1)$$

where $s \in \{4, 8, 16\}$ with $(H_s, W_s) \in \{(96, 96), (48, 48), (24, 24)\}$. Each feature map is reshaped into $N_s = H_s \times W_s$ tokens $\mathbf{x}_i \in \mathbb{R}^C$.

B Router. For each scale, a depthwise 3×3 convolution captures local spatial context from the feature map. The router input concatenates each token with its local context:

$$\mathbf{r}_i = [\mathbf{x}_i \mid \text{DWConv}_{3 \times 3}(\mathbf{X}_s)_i] \in \mathbb{R}^{2C}, \quad (2)$$

where $\mid$ denotes channel-wise concatenation, yielding a $2C$ -dimensional vector that encodes both the token's content and its spatial neighborhood. A two-layer MLP produces routing logits:

$$\mathbf{h}_i = W_2 \cdot \text{GELU}(W_1 \mathbf{r}_i + b_1) + b_2 \in \mathbb{R}^E, \quad (3)$$

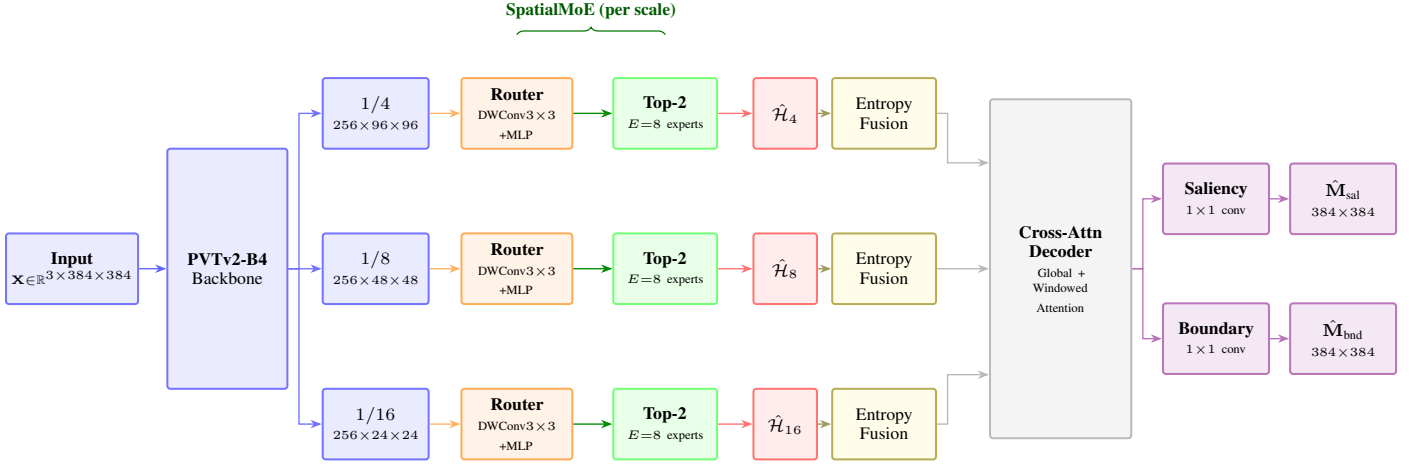


Figure 1. SpatialMoE-SOD architecture. Input $\mathbf{X} \in \mathbb{R}^{3 \times 384 \times 384}$ is processed by a PVTv2-B4 backbone to produce multi-scale features at 1/4, 1/8, 1/16 resolution. Each scale is independently processed by a SpatialMoE layer: a depthwise convolutional router selects top- $K=2$ experts from 8 via noisy gating, and a per-token entropy map $\hat{\mathcal{H}}$ is computed. The decoder merges scales via global and windowed cross-attention, with entropy features injected via learned additive fusion at each scale. Two 1×1 conv heads produce saliency and boundary logits at full resolution.

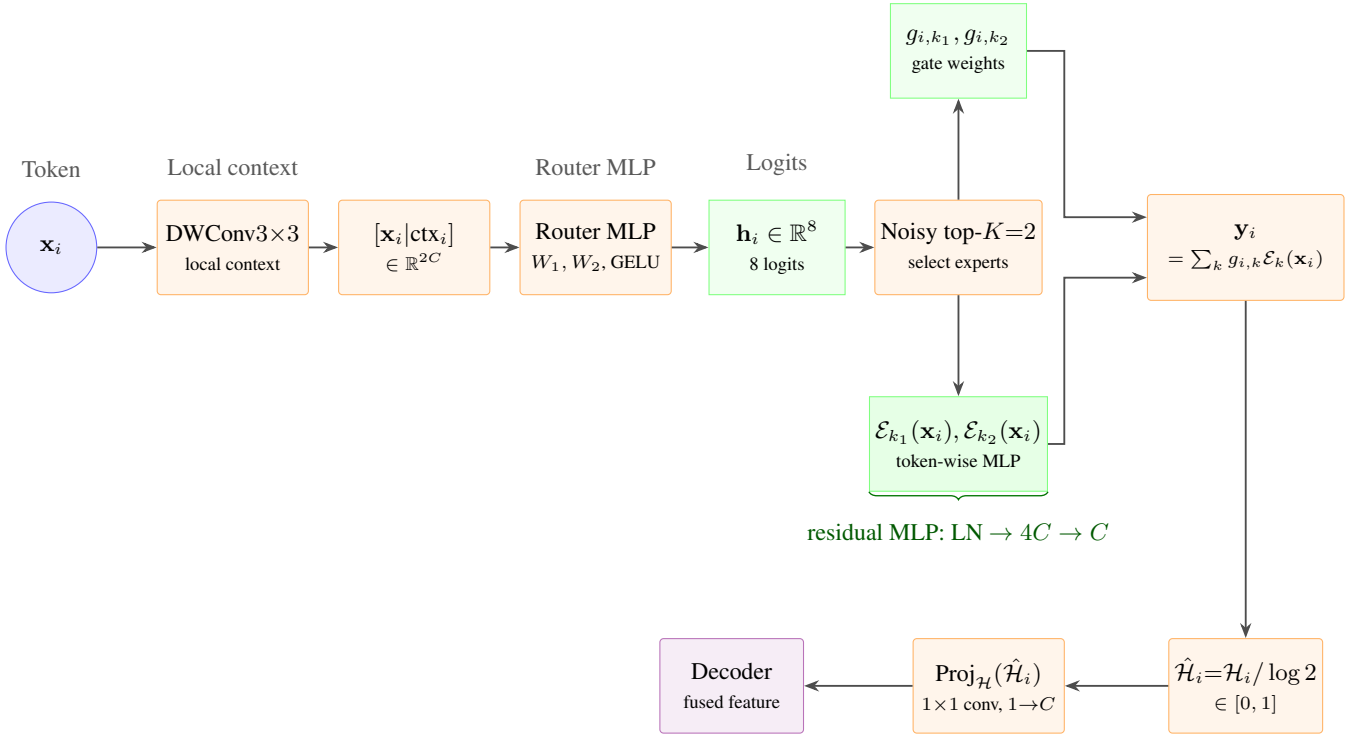


Figure 2. Routing mechanism for token $\mathbf{x}_i$ at one pyramid scale. A depthwise 3×3 convolution extracts local spatial context; the concatenation $\mathbf{r}_i = [\mathbf{x}_i | \text{ctx}_i] \in \mathbb{R}^{2C}$ is passed through a two-layer MLP to produce 8 routing logits. Noisy top- $K=2$ selection (Gaussian noise during training) picks two experts, branching into gate weights (top) and expert outputs (bottom); gate weights are softmax over selected logits only. The selected experts process the token independently; their outputs are summed with gate weights. Routing entropy $\hat{\mathcal{H}}_i \in [0, 1]$ measures assignment uncertainty and is injected as a feature channel into the decoder via a learned 1×1 projection.

with $W_1 \in \mathbb{R}^{128 \times 2C}$, $W_2 \in \mathbb{R}^{E \times 128}$, $E=8$ experts. The hidden dimension 128 provides a bottleneck between the $2C=512$ input and $E=8$ output.

C Noisy Top-K Routing. During training, Gaussian noise is added with input-dependent scale to encourage exploration:

$$\tilde{h}_{i,j} = h_{i,j} + \epsilon_{i,j}, \quad \epsilon_{i,j} \sim \mathcal{N}(0, \sigma_j^2(\mathbf{x}_i)), \quad (4)$$

where $\sigma_j(\mathbf{x}_i) = \text{softplus}(w_j^\top \mathbf{x}_i + b_j)$ is a learned per-expert, per-token noise scale. At inference, noise is disabled ($\epsilon=0$). Top- $K=2$ experts are selected per token; gate weights are softmax over selected logits only:

$$g_{i,k} = \frac{\exp(\tilde{h}_{i,k})}{\sum_{j \in \mathcal{T}_i} \exp(\tilde{h}_{i,j})}, \quad k \in \mathcal{T}_i = \text{top-k}(\tilde{h}_{i,j}, K=2). \quad (5)$$

This produces a sparse assignment: each token attends to exactly 2 of 8 experts, and each expert processes only the tokens that selected it.

D Sparse Expert Aggregation. Each expert $\mathcal{E}_k$ executes only on tokens that selected it. For expert k , let $\mathcal{A}_k = \{i : k \in \mathcal{T}_i\}$ denote its assigned tokens:

$$\mathbf{y}_i = \sum_{k \in \mathcal{T}_i} g_{i,k} \cdot \mathcal{E}_k(\mathbf{x}_i), \quad (6)$$

implemented via a loop over experts with mask-based selection (true sparsity, not dense with masking). Each expert is a token-wise residual MLP:

$$\mathcal{E}_k(\mathbf{x}) = \mathbf{x} + W_k^{(2)} \cdot \text{GELU}(W_k^{(1)} \cdot \text{LN}(\mathbf{x})), \quad (7)$$

with $W_k^{(1)} \in \mathbb{R}^{4C \times C}$, $W_k^{(2)} \in \mathbb{R}^{C \times 4C}$ ($4\times$ expansion from $C=256$ to $4C=1024$). Each scale has 8 independent experts with no weight sharing (24 total). Reshaping per-token outputs yields $\mathbf{Y}_s \in \mathbb{R}^{C \times H_s \times W_s}$.

E Routing Entropy. Per-token routing entropy measures assignment uncertainty over the selected experts:

$$\mathcal{H}_i = - \sum_{k \in \mathcal{T}_i} g_{i,k} \log g_{i,k}, \quad \hat{\mathcal{H}}_i = \mathcal{H}_i / \log 2 \in [0, 1]. \quad (8)$$

Division by $\log 2$ normalizes to $[0, 1]$, where 0 indicates a single expert dominates and 1 indicates uniform assignment. The normalized entropy map $\hat{\mathcal{H}}_s \in \mathbb{R}^{1 \times H_s \times W_s}$ is fused with MoE output via a learned additive block:

$$\mathbf{F}_s = \text{Proj}_Y(\mathbf{Y}_s) + \lambda_s \cdot \text{Proj}_{\mathcal{H}}(\hat{\mathcal{H}}_s), \quad (9)$$

where $\text{Proj}_{\mathcal{H}}$ is a 1×1 convolution from 1 to C channels and λ_s is a learnable scalar initialized to 0.1. This allows the decoder to condition its attention on routing confidence without requiring the entropy to be differentiable through the expert weights.

F Decoder. The decoder merges coarse-to-fine features via top-down cross-attention. At the coarsest scale, $\mathbf{F}_{16}$ is upsampled and used as queries with $\mathbf{F}_8$ as keys and values (8-head global attention):

$$\mathbf{F}'_8 = \text{GlobalCrossAttn}(Q = \text{Up}(\mathbf{F}_{16}), KV = \mathbf{F}_8). \quad (10)$$

Global context is then injected into the fine scale: $\mathbf{F}_4^{\text{ctx}} = \mathbf{F}_4 + \text{Proj}_g(\text{AvgPool}(\mathbf{F}'_8))$. $\mathbf{F}'_8$ is upsampled and used as queries with $\mathbf{F}_4^{\text{ctx}}$ as keys and values using 7×7 windowed attention for efficiency:

$$\mathbf{F}'_4 = \text{WinCrossAttn}(Q = \text{Up}(\mathbf{F}'_8), KV = \mathbf{F}_4^{\text{ctx}}). \quad (11)$$

Three refinement blocks (each: $\text{Conv } 3 \times 3 \rightarrow \text{GroupNorm} \rightarrow \text{ReLU} \rightarrow \text{Conv } 3 \times 3 \rightarrow \text{GroupNorm} \rightarrow \text{ReLU} \rightarrow \text{residual addition}$) progressively upsample from $1/4$ to full resolution. Two 1×1 convolution heads produce saliency logits $\hat{\mathbf{M}}_{\text{sal}}$ and boundary logits $\hat{\mathbf{M}}_{\text{bnd}}$ at 384×384 .

Table 1. Global performance on WXSOD test splits.

Set	MAE $\downarrow$	S_ϕ $\uparrow$	F_β^{max} $\uparrow$	E_ϕ^{adp} $\uparrow$
Synth.	0.0192	0.9139	0.9015	0.9591
Real	0.0168	0.9151	0.8936	0.9530

G Training Objective. The total loss combines four active terms (all other terms in the codebase are inactive):

$$\mathcal{L} = \mathcal{L}_{\text{BCE}} + \lambda_{\text{IoU}} \mathcal{L}_{\text{IoU}} + \lambda_{\text{LB}} \mathcal{L}_{\text{LB}} + \lambda_{\text{IMP}} \mathcal{L}_{\text{IMP}}, \quad (12)$$

with $\lambda_{\text{IoU}}=1.0$, $\lambda_{\text{LB}}=\lambda_{\text{IMP}}=0.01$. $\mathcal{L}_{\text{BCE}}$ is binary cross-entropy with logits between predicted saliency logits and ground-truth mask. $\mathcal{L}_{\text{IoU}}$ is the soft IoU loss: $1 - \text{mean}_b \left[\frac{\sum_{hw} \sigma(\hat{M}_b) \cdot M_b + \epsilon}{\sum_{hw} \sigma(\hat{M}_b) + \sum_{hw} M_b - \sum_{hw} \sigma(\hat{M}_b) \cdot M_b + \epsilon} \right]$. $\mathcal{L}_{\text{LB}}$ is the load-balancing loss: $\frac{1}{|\mathcal{S}|} \sum_s E \sum_j f_j^{(s)} P_j^{(s)}$, where f_j is the fraction of all routing slots assigned to expert j , computed from hard assignments and detached, and P_j is the differentiable mean gate mass. $\mathcal{L}_{\text{IMP}}$ is the importance loss: $\frac{1}{|\mathcal{S}|} \sum_s (\text{std}(\mathbf{P}^{(s)}) / \text{mean}(\mathbf{P}^{(s)}))^2$, where $\mathbf{P}^{(s)} \in \mathbb{R}^E$ denotes the per-expert mean gate-mass vector at scale s . Deep supervision, boundary, and SSIM losses are defined in the codebase but inactive in the reported configuration; deep supervision is disabled at the model level, while the boundary and SSIM loss weights are zero.

IV. EXPERIMENTS

A Setup. We evaluate on WXSOD [3], which provides 12,891 training images (scene-aware 80/20 split), 1,500 synthetic test images across 9 weather categories, and 554 real-world test images across 5 categories (fog, rain, snow, dark, low-light). Images are aspect-preserving resized to 384×384 ; augmentations include horizontal flip ($p=0.5$), color jitter ($p=0.5$), and ImageNet normalization. Training uses DDP on 2 GPUs, AMP (FP16), AdamW optimizer (lr 10^{-4} for newly introduced modules and 2×10^{-5} for the backbone, weight decay 10^{-4}), WarmupCosine schedule (1% warmup, 50 epochs), gradient clipping at 1.0, effective batch size 32. The backbone is frozen during the first epoch and unfrozen from epoch 1 onward. Metrics: MAE, S_ϕ [6], E_ϕ^{adp} [7], F_β^{max} [8]. The model has 66.27M parameters and 278.2G MACs per 384×384 input image.

B Main Results. Table 1 reports global performance on both test splits. The model achieves MAE of 0.0192 on synthetic data and 0.0168 on real-world data. Real-world MAE (0.0168) is slightly lower than synthetic MAE (0.0192). However, this comparison is confounded by test-set composition: the synthetic test set includes compound weather conditions (rain+fog, rain+snow, snow+fog) that are absent from the real-world test set, which may increase synthetic-test error independently of any domain-shift effect.

C Weather-Wise Analysis. Table 2 reports per-weather performance on real-world test. Performance varies by $2.6\times$ across weather types: snow yields the lowest MAE (0.0094, $N=90$), while low-light yields the highest (0.0243, $N=93$). Fog (0.0148,

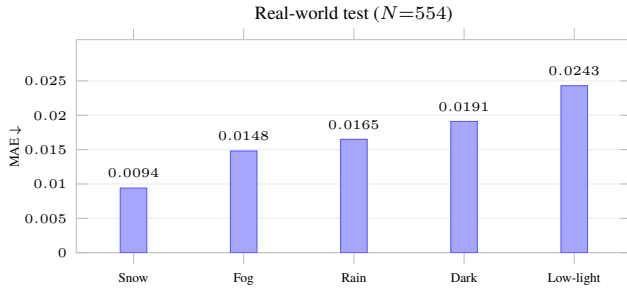


Figure 3. Weather-wise MAE on real-world test. Snow yields the lowest error, while low-light yields the highest.

Table 2. Weather-wise results on real-world test (554 images).

Weather	N	MAE ↓	S_ϕ ↑	$F_\beta^{\max}$ ↑
Snow	90	0.0094	0.9478	0.9421
Fog	126	0.0148	0.9179	0.8914
Rain	120	0.0165	0.9157	0.8935
Dark	125	0.0191	0.9072	0.8885
Low-light	93	0.0243	0.8897	0.8629

$N=126$), rain (0.0165, $N=120$), and dark (0.0191, $N=125$) fall between these extremes. On synthetic data, compound weather conditions (rain+fog) yield the worst MAE (0.0237), suggesting that compositing multiple degradation types compounds difficulty.

D Routing Entropy. Table 3 reports mean normalized entropy per scale. Mean normalized entropy ranges from 0.68 to 0.69 across scales, indicating moderate routing uncertainty. Scale 1/16 exhibits marginally lower entropy (0.680–0.682) than scales 1/4 (0.691) and 1/8 (0.693), although the difference is small. The difference between synthetic and real entropy is less than 0.003 at all scales, indicating similar mean routing entropy across the two test splits.

Table 3. Mean routing entropy (normalized by log 2).

Scale	Synthetic	Real
1/4	0.6910	0.6912
1/8	0.6931	0.6931
1/16	0.6800	0.6822

E Forced-Expert Analysis. To probe whether routing provides meaningful specialization, we forced all tokens at scale 1/4 to expert 0 (bypassing the router), while scales 1/8 and 1/16 remained routed normally. On real-world test: MAE increased by +0.0002 (+0.9%), S_ϕ decreased by −0.0012 (−0.1%). This marginal degradation admits two interpretations: (a) the 8-expert pool is over-parameterized for the task, and a single expert can handle most tokens; or (b) experts have learned similar representations, so bypassing routing has limited impact. Since only 1 of 3 scales was affected, the full effect of disabling all routing remains unknown. This result should not be interpreted as evidence for or against routing utility; a complete ablation disabling all three routers simultaneously would be needed.

F Discussion. The model demonstrates that spatial token-level routing is feasible for single-RGB SOD without weather-specific supervision. However, several important limitations must be noted. First, no controlled comparison against existing methods (NIFM) has been performed; the reported results establish a baseline but do not demonstrate state-of-the-art performance. Second, the model was trained once without replicate runs, so variance estimates are unavailable. Third, the forced-expert analysis suggests that expert specialization may be limited, and the benefit of multi-expert routing over a single larger expert remains unclear. Fourth, the routing entropy values are moderate but their relationship to prediction quality is correlational—we have not established that high-entropy tokens are systematically harder or that the entropy feature improves decoder performance compared to a baseline without it.

V. CONCLUSION

We presented SpatialMoE-SOD, which routes individual spatial tokens to separate expert networks at three independent pyramid scales via context-aware routers, without any weather-specific supervision. Per-token routing entropy is normalized and injected as an explicit decoder feature via learnable additive fusion. On WXSOD, the model achieves MAE 0.0192 (synthetic, 1,500 images) and 0.0168 (real-world, 554 images), with weather-wise MAE ranging from 0.0094 (snow) to 0.0243 (low-light).

Several limitations warrant future investigation. The model was trained once without replicate runs, so variance estimates are unavailable. No controlled comparison against existing methods (NIFM [1]) has been performed. The forced-expert analysis suggests limited differentiation among experts at scale 1/4, and the contribution of the entropy fusion mechanism relative to a baseline without it has not been ablated. The routing entropy values are moderate (0.68–0.69) but their causal relationship to prediction quality remains unestablished. Future work should complete the ablation matrix (routing entropy, number of experts, K value, scale independence), perform head-to-head comparison with existing methods, and analyze per-expert feature specialization via activation visualization.

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Figure 4. Qualitative results on real-world WXSOD test. Three examples spanning snow, fog, and dark conditions. The error map (rightmost column, red) highlights spatial regions where the prediction deviates from ground truth; larger errors concentrate at object boundaries and in regions with degraded visibility.